

Question	Marking Guidance	Mark	Comments
06.1	1. (μg because) very little ammonia (in soil); 2. (μg because) avoids use of (lots of) decimal places (in their results) / avoids the use of powers of 10 / avoids the use of standard form; 3. (g^{-1}) to allow comparisons (between samples);	2 max	2. Accept makes numbers more manageable 2. Accept makes easier to plot graph
06.2	Answer between 4.5 and 4.6 $\mu\text{g g}^{-1} \text{ day}^{-1}$;; Award 1 mark for correct number but wrong / no units	2	Ignore plus or minus signs Accept 'per gram' AND / OR 'per day'
06.3	1. pH 4.3 / B has fastest rate of breakdown (of ammonia); 2. A + B / mixture at pH 6.9 slowest / slower (than A or B); 3. Suggests (community / bacteria at) pH 4.3 / B doesn't work (well) at pH 6.9 / pH of mixture;	3	2. Not just ref. to A and then B on their own 3. Accept converse that only (community / bacteria at) pH 6.9 / A is working in the mixture
06.4	(Species S because) no mark 1. Species S change of 990,000 (per gram of soil); 2. Species T change of 9,900 (per gram of soil); OR (Species T because) no mark 3. Species S has 99% change; 4. Species T has 9900% change;	2	Award MP1 and 2 OR MP3 and 4 1. and 2. Accept standard forms 9.9×10^5 and 9.9×10^3 for either Accept for 1 mark for 100 times greater in correct context with no other calculations shown

06.5	1. They didn't count bacteria / cells / population(s);	4	Ignore ref. to other
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